

# VYSHAK V

## AI ENGINEER

LLM Systems • Agentic AI • RAG • Computer Vision • Full-Stack

Chennai, India | +91 82203 73329 | vyshak1705@gmail.com | [linkedin/vyshak-v](https://www.linkedin.com/in/vyshak-v) | Portfolio

### PROFESSIONAL SUMMARY

AI Engineer with 1.5+ years of experience building production AI systems at IQ TechMax, including real-time multilingual conversational agents, an enterprise AI concierge for a Dubai client, and a computer-vision inspection system for Magna. Specialised in LLM engineering (RAG, agentic workflows, function calling, agent memory and cost-aware SLM/LLM routing), with full-stack delivery across Python/FastAPI, Node.js, React, PostgreSQL/pgvector, Redis and Docker-based deployment.

### TECHNICAL SKILLS

**Programming & Frameworks:** Python (FastAPI, Pydantic, NumPy), TypeScript (React, Next.js, Node.js)

**LLMs & Agents:** RAG, Agentic Workflows, Function Calling, Structured Outputs, Agent Memory, Context & Prompt Engineering, SLM/LLM Routing, Semantic Caching, LLM Evaluation, LoRA

**Computer Vision:** OpenCV, YOLO, ArUco Pose Estimation, Image-Based Measurement

**Full-Stack Development:** REST APIs, SSE Streaming, SQLAlchemy, Alembic, Celery, Tailwind CSS

**MLOps & Deployment:** Model Deployment (FastAPI + Docker), Docker Compose, CI/CD (GitHub Actions), Containerized Deployments, iOS Deployment (Xcode, TestFlight)

**Databases:** PostgreSQL, Redis, Vector Database (pgvector), Object Storage (MinIO)

### PROFESSIONAL EXPERIENCE

#### AI Engineer - IQ TechMax IT Services and Consulting, Chennai

Feb 2025 – Present

##### KASPAR - Enterprise AI Concierge & Guest-Intelligence Platform · Client Project, Dubai

Stack: Python, FastAPI, PostgreSQL/pgvector, Redis, PyTorch, Node.js/Express, React, Docker

- Engineering the AI layer of a VIP delegation-management platform: a RAG-based AI Concierge and AI Planner that convert natural-language requests into structured intents and versioned itineraries, with persistent session context.
- Built the end-to-end knowledge pipeline (document parsing, metadata extraction, semantic chunking, embeddings, pgvector storage) with tier-aware retrieval and access isolation for guest-specific knowledge.
- Designed a multi-layer memory architecture across guest, planner, supplier and delegation context for personalised interactions across sessions and events.
- Developing a hybrid SLM/LLM inference stack with self-hosted models, Redis semantic caching, response-quality validation, automatic escalation to larger models and an evaluation framework, LoRA fine-tuning in progress.
- Building a Recommendation Engine for similar-delegation and planner-preference suggestions, integrated AI services with the React admin portal and Node.js API on Docker.

##### AI-Driven Robotic Arm Inspection · Client Project, Magna

Stack: Python, OpenCV, YOLO, ArUco, NumPy, Unity 6, Xcode/TestFlight

- Developed a computer-vision service that compares robotic torch-wire geometry against CAD references using YOLO detection, ArUco marker tracking, camera pose estimation and contour/line fitting.
- Engineered measurement pipelines for wire length, tilt, angular and spatial deviation, aggregating frame-level results for consistency, set up a controlled capture rig for repeatable calibration.
- Integrated the service with a Unity 6 iPad app for CAD alignment and real-time visualisation, distributed via TestFlight, with the CV engine decoupled as an independent service.

##### IQ Agent - Interactive Real-Time AI Agents · Company Product,

- Developed AI-agent capabilities for a real-time conversational platform supporting chat, voice and video interactions across multilingual and industry-specific use cases.
- Built an HR assessment agent using function calling, structured outputs and context engineering for multi-step workflow automation.
- Engineered core components (intent detection, context management, model integration and Pinecone-based RAG) and a customisable assistant with web search powering personalised dashboards.

### EDUCATION

B.E., Computer Science and Engineering - KPR Institute of Engineering and Technology, Coimbatore

2021 – 2025